

Detecting Outliers in Financial Transactions

A Data Analytics Approach for Identifying Unusual Financial Transactions

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Financial Transaction Analysis

Financial institutions process thousands of transactions daily. Most follow normal patterns — but some transactions carry unusually high or low values, known as **outliers**.

Outliers Can Occur Due To:

- Fraud
- Data entry errors
- System issues
- Rare but legitimate transactions

The Challenge

Identifying these transactions manually becomes increasingly difficult as transaction volume grows. Organizations need a **data-driven approach** to detect unusual transactions and support timely investigation.



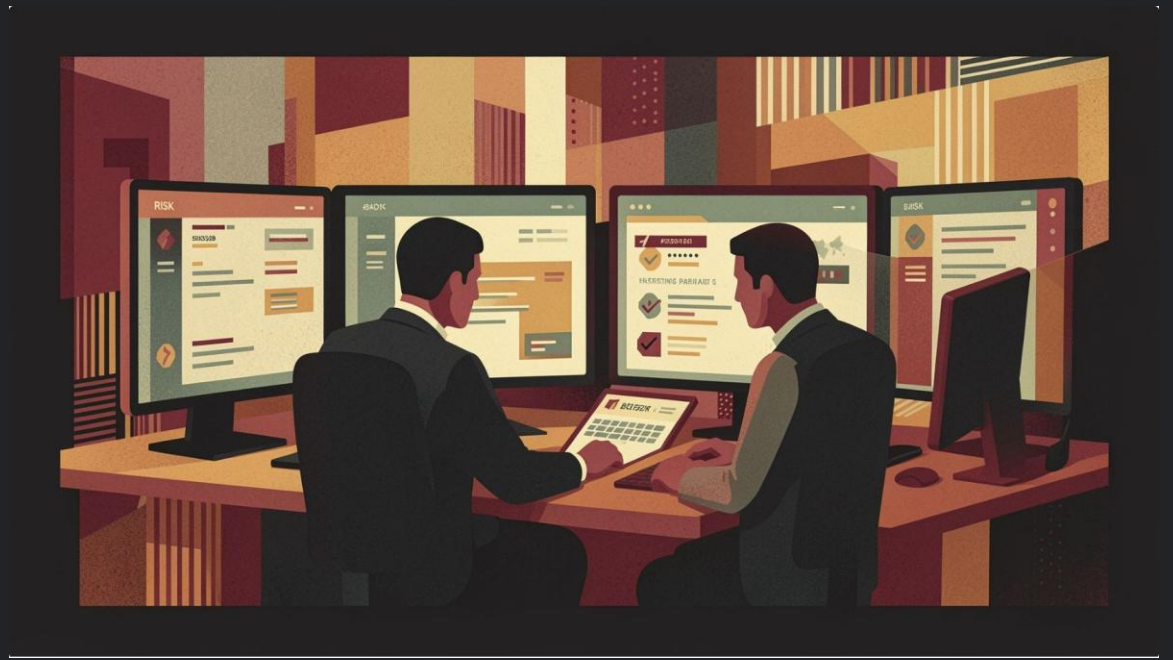
The Business Problem

Problem Statement

A bank processes a large number of financial transactions every day. Some transactions may differ significantly from normal behaviour. These unusual transactions may:

- Indicate potential fraudulent activity
- Result from data entry or system errors
- Affect financial analysis and forecasting
- Increase financial and operational risk
- Require manual investigation

⚠ Traditional rule-based methods may miss important anomalies or generate too many false alerts.



Objective

To identify abnormal transaction patterns, reduce unnecessary alerts, and provide actionable information for risk awareness and decision-making.

Analytical & Business Solutions

Five key solutions to address outlier detection and transaction risk management:

1

Investigate High-Risk Transactions

- Review transactions identified as highly unusual
- Verify with customer or department
- Escalate confirmed suspicious transactions

2

Strengthen Fraud Prevention

- Introduce additional verification for high-risk transactions
- Set suitable transaction limits
- Improve fraud prevention controls

3

Correct Data & System Errors

- Investigate outliers caused by incorrect data
- Correct inaccurate records
- Improve data validation and system controls

4

Improve Risk Monitoring & Alerts

- Establish automated alerts for unusual transactions
- Prioritize high-risk transactions
- Maintain investigation records

5

Act on Legitimate Outliers

- Identify unusual transactions that are legitimate
- Update customer transaction profiles
- Reduce repeated false alerts

Solution Implementation

How the organization implements solutions through a structured, end-to-end process:

Automated Transaction Monitoring

Connect transaction data to a monitoring system. Automatically check new transactions against normal patterns and generate alerts for unusual activity.

Risk-Based Investigation

Classify alerts as Low, Medium, or High Risk. Review medium-risk transactions and immediately investigate high-risk transactions.

Additional Verification

Request additional customer authentication when required. Verify suspicious transaction details. Hold or restrict transactions until verification is complete.

Data & System Controls

Introduce validation checks during transaction entry. Prevent invalid amounts and dates. Maintain accurate customer transaction records.

Continuous Monitoring

Review previous alerts and investigation results. Identify false alerts, improve thresholds, and continuously update the system based on changing transaction behaviour.



Dataset & Methodology

Dataset Overview

A financial transaction dataset was created for this analysis, with intentionally introduced data-quality problems to demonstrate the full cleaning process.

4,080

Total Rows

Transaction records

14

Total Columns

Data attributes

Data Specifications

- **Format:** CSV
- **Data Type:** Financial Transaction Data
- **Purpose:** Data cleaning, exploratory analysis, and outlier detection
- **Cleaning Tool:** Python / Pandas

Financial Transaction Dataset

The dataset contains **14 attributes** describing each financial transaction:

s.no	Field	Description
1	Transaction_ID	Unique identification number
2	Customer_ID	Identifies the customer
3	Customer_Name	Name assigned to the customer
4	Account_Number	Customer's account identifier
5	Transaction_Date	Date the transaction occurred
6	Transaction_Time	Time the transaction occurred
7	Transaction_Type	Debit, credit, transfer, or withdrawal

s.no	Field	Description
8	Transaction_Amount	Monetary value of the transaction
9	Merchant_Name	Merchant associated with the transaction
10	Payment_Method	UPI, card, NEFT, etc.
11	City	Location of the transaction
12	State	State associated with the transaction
13	Device_Type	Device or channel used
14	Transaction_Status	Successful, failed, or pending

Data Cleaning

Why Cleaning is Necessary: The raw dataset contains errors and inconsistencies. Cleaning is essential before performing outlier detection — poor-quality data produces misleading results.

1 Identify Missing Values

Detect null or empty fields in the dataset

2 Handle Missing Values

Fill or remove records with missing data appropriately

3 Remove Duplicates

Eliminate exact duplicate transaction entries

4 Check Duplicate IDs

Ensure each transaction has a unique identifier

5 Correct Invalid Dates

Fix date format errors and impossible date values

6 Fix Negative Amounts

Validate that amounts are positive or correctly signed

7 Standardize Text

Ensure consistent formatting across all text fields

8 Remove Extra Spaces

Trim leading and trailing whitespace throughout

9 Correct Spelling Errors

Fix misspelled merchant and customer names

10 Validate Status Values

Ensure status values are from the approved list only

11 Validate Data Types

Confirm each column has the correct data type

12 Validate Outlier Values

Check for extreme values that may be data errors

Data Quality Transformation

Before Cleaning

- Missing values
- Duplicate transactions
- Invalid dates
- Negative amounts
- Spelling mistakes
- Extra spaces
- Inconsistent values
- Incorrect data formats

After Cleaning

- Consistent data
- Valid transaction records
- Correct formats
- Reduced duplication
- Fewer data-quality errors
- Reliable dataset for analysis



Raw Dataset | Cleaning | Validation | Reliable Dataset

> *"Clean data is essential because outlier detection performed on incorrect data may identify data errors instead of genuine unusual transactions."*



Expected Business Impact & Conclusion

The Challenge

Large transaction volumes make manual outlier detection difficult. Outliers may result from fraud, errors, system issues, or legitimate activity.

The Solution

A structured data-driven approach helps identify and investigate unusual transactions while improving risk monitoring, fraud prevention, data quality, and decision-making.

Key Takeaway

Outliers are not automatically fraud — they are transactions that require further investigation. Continuous monitoring reduces false alerts and adapts to changing patterns.